

Jake Kim

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EDUCATION

Purdue University

B.S in Mechanical Engineering

West Lafayette, IN

May 2026

PROFESSIONAL EXPERIENCE

Honeywell | Advanced Connected and Sustainability Technologies

Jun 2025 – Aug 2025

Materials Engineering Intern

Plymouth, MN

- Led scrap **silver** powder reusability investigation across multiple sintering reuse cycles; delivered **33%** reduction in material costs through powder metallurgy and materials testing.
- Developed sinter manufacturing protocols and led ion mill process optimization using ANSYS DEM simulations and process mapping, improving throughput.
- Spearheaded AM evaluation initiative — assessed technologies and designed validation test articles for deployment across MN sites.

Manufacturing and Materials Research Laboratories

May. 2024 – May 2026

Undergraduate Research Fellow

West Lafayette, IN

- Executed **41** GPU-accelerated DEM simulations modeling metal powder spreading in LPBF, reducing compute time **90%** via **NVIDIA V100/A100** acceleration across **20,000** data points.
- Built multivariate regression framework (**$R^2=0.978$**) linking particle-scale parameters to powder bed quality; established defect likelihood criteria for cantilevered LPBF structures.

Purdue Center for Particulate Products and Processes

May 2023 – May 2024

Undergraduate Researcher

West Lafayette, IN

- Performed XRCT imaging of corn kernels, generating **100+** high-resolution 3D models for integration into Discrete Element Modeling simulations and establishing repeatable imaging workflow for research applications.
- Modeled axial stresses in the plug section of a screw feeder using Discrete Element Modeling (DEM) to optimize particle flow and reduce mechanical errors.

PUBLICATIONS AND PRESENTATIONS

- Kim, J.**, Khattab, D.W., Titus, M.S., & Mort, P. Multivariate Modeling of Powder Spreading in Laser Powder Bed Fusion. *Powder Technology*, 477 (2026), 122553 <https://doi.org/10.1016/j.powtec.2026.122453>
- Khattab, D.W.S., Shen, X., **Kim, J.**, et al. (2025). Powder flowability of elemental niobium, C103, and other structural materials for powder bed fusion. *TMS Annual Meeting & Exhibition*, Las Vegas, NV.
- Kim, J.**, Khattab, D.W.S., Shen, X., Titus, M.S., & Mort, P. (2024). Impact of Boundary Layer Rheology, Particle Morphology, and Surface Energy on Powder Bed Quality in Laser Powder Bed Fusion. *SURF Symposium*, Purdue University.
- Mort, P., Stumpf, D., Varma, R., Khattab, D., Shen, X., **Kim, J.**, Titus, M., Sealy, M., Jaworek, J., & Feltner, L. (2026). Powder metal characterization for additive manufacturing. In *XMO Conference: Manufacturing at the Crossroads of America*.

TECHNICAL PROJECTS

Variable Stiffness Robotic Gripper | Manufacturing and Design Lead

Jan 2026 – May 2026

- Built a non-Newtonian fluid-based robotic gripper achieving **139%** stiffness increase (**0.520 N/mm** above baseline) and **6 N** grip force
- Fabricated 4-bar linkage Gripper and Fluid module via additive manufacturing; characterized PEG/silica shear thickening fluid to optimize performance

Bioinspired L-PBF Lattice for Bending Optimization | Design Lead

Aug 2025 – Dec 2025

- Fabricated a bamboo-inspired hexagonal lattice in **316L** stainless steel via Renishaw LPBF, applying CAD tools and process optimization to mimic bamboo's graded architecture; achieved **20% weight reduction** with **2:1** strength tradeoff.

LEADERSHIP

Mechanical Engineering Sophomore Seminar

Aug 2025 – Dec 2025

Mechanical Engineering Mentor

West Lafayette, IN

- Mentor and support over **40** sophomore Mechanical Engineering students, fostering academic success and community engagement.
- Cultivated communication and interpersonal skills through collaboration with peers, faculty, staff, and alumni.

Purdue Recreation and Wellness

May 2022 – May 2026

Member Services Manager

West Lafayette, IN

- Manage and coordinate front-line operations for one of the largest university recreation centers in the U.S., serving **5000 +** daily visitors, ensuring smooth and efficient operation of member services during operating hours.

Cornerstone Christian Fellowship

Aug 2025 – May 2026

Coffee Ministry Lead

West Lafayette, IN

- Lead the Coffee Ministry for one of Purdue's largest Asian American Christian Fellowships, conducting morning Bible studies and fostering community connection.

TECHNICAL SKILLS

Software Tools: JMP, Altair EDEM, Ansys, Solidworks, Fusion 360, Creo, Autodesk Inventor

Languages: Python, Matlab, Simulink

Materials Characterization: SEM, X-ray CT, Ion Mill, BET, Eddy Current, Sintering, Granudrum, 3D printing